A* Examination Guide

Built from 16 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the Paper 1 question papers and mark schemes. Model answers use the marking points examiners list, and every warning comes from a “reject”, “ignore”, “allow” or “max” note in the mark schemes. Paper 1 tests any part of the specification, so learn the content and the answer patterns together.

1 How Paper 1 is built
2 Twenty mark-scheme rules
3 Cells, biological molecules and enzymes
4 Nutrition, digestion and food tests
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Papers analysed: Jan 2019 (legacy 4HB0), Jun 2019, Jan 2020, Nov 2020, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 HOW PAPER 1 IS BUILT

Format: 1 hour 45 minutes, 90 marks, 7–9 structured questions (the legacy 4HB0 paper was 2 hours). That is just over 1 minute per mark. Questions mix recall, a few multiple-choice items, a graph to plot, a calculation, a 4–7 mark method and 3–6 mark explanations.

Topic	4HB1 papers (of 15)	What is asked most
Calculations	15	magnification, percentage change, ratio, mean, rate, standard form, energy
Investigations and methods (4–7 marks)	15	enzymes, food energy, breathing, hearing, pupil, disinfectants
DNA, RNA, cell division, mutation	11	structure, base pairing, DNA vs RNA, mitosis vs meiosis
Blood components, clotting, blood groups	8	match cell to function, clotting cascade, agglutination
Heart, vessels, tissue fluid	8	chambers and vessels, valves, pressure changes, tissue fluid formation
Homeostasis (temperature, glucose)	8	hypothalamus, vasodilation, sweating, insulin and glucagon
Enzymes	8	temperature/pH effects, active site, immobilised enzymes, rate from a graph
Nervous system	7	neurone types, reflex arc, synapse (6 marks)
Kidney and ADH	7	ultrafiltration, reabsorption, ADH, dialysis, alcohol
Breathing and gas exchange	7	inhalation mechanism, alveoli adaptations, lung volumes
Inheritance	7	blood groups, sex-linked conditions, pedigrees, probability

Topic	4HB1 papers (of 15)	What is asked most
Diet, food tests, food energy	7	Benedict's, biuret, iodine, emulsion; burning food
Skeleton, joints, muscles	7	synovial joint labels, antagonistic muscles, muscle types
Pathogens, immunity, antibiotics	7	vaccination, phagocytosis, resistance, aseptic technique
Eye and ear	6	accommodation, pupil reflex, defects, hearing tests, balance
Reproduction	6	menstrual cycle hormones, sperm structure, meiosis, vasectomy, breastfeeding

2 TWENTY MARK-SCHEME RULES

1. **Never write "amount":** say mass, volume or concentration. "Ignore amount" and "reject amount" appear in most control-variable questions.
2. **Methods need detail:** what you measure, how, the control variables, repeats and a mean. One mark per creditworthy step, so write 6–7 steps for a 6-mark method.
3. **Food tests need the full procedure:** Benedict's must be **heated in a water bath**; give the colour change from start to end (blue → brick red).
4. **Enzymes are denatured, not killed:** explain via the **active site** changing shape so the substrate no longer fits and fewer **enzyme–substrate complexes** form.
5. **Osmosis uses water potential:** "ignore reference to high/low concentration of water" (Nov 21).
6. **Graphs:** label both axes with units, the independent variable on x, a scale using at least half the grid, accurate plots, lines as instructed ("join with straight lines"), and a key for two lines. Bar charts for categories, with bars of equal width.
7. **Describe a trend:** state the direction, quote figures with units, and note changes in rate or exceptions.
8. **Compare:** use comparative words (higher, faster) and refer to both items. Reverse arguments are allowed.
9. **Calculations:** show working (error carried forward is allowed); convert units first (1 mm = 1000 µm); give the decimal places asked (172.1 not 172, Jun 19).
10. **Genetic diagrams:** parent genotypes, gametes, offspring genotypes, phenotypes and the probability. Use a capital letter for the dominant allele; for sex-linked alleles write $X^{D}X^{d}$.
11. **Be precise with names:** ligaments join bone to bone; tendons join muscle to bone. Left versus right chambers: "maximum of one mark if right atrium/ventricle given" (Jan 20).
12. **Antibodies not "fighting":** "ignore fight infection"; say phagocytosis or antibody production.
13. **Clots, not scabs:** "R scab" (Nov 23). Say clot, plug or mesh of fibrin.
14. **Inhalation:** diaphragm contracts and **flattens**; "reject moves down"; "ignore external/internal" for intercostals (Jun 25).
15. **Respiration equation:** $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ energy/ATP). Balanced symbols required (Jun 25).
16. **Labelling:** label lines must touch the structure; extra labels beyond the number asked lose marks.
17. **Multiple choice:** one letter only; the mark scheme explains why each other option is wrong, so reason it through.
18. **Count the marks:** give one extra creditworthy point for "any four from" questions; the maximum is capped.
19. **Link cause and effect** in "explain": each statement followed by "so" or "because".
20. **Safety precautions need a reason** ("max if no explanation given"): goggles because hot water may spit.

3 CELLS, BIOLOGICAL MOLECULES AND ENZYMES

Structure	Function (as credited)
nucleus	contains chromosomes/DNA; controls the cell
cell membrane	controls which substances enter and leave
mitochondria	site of aerobic respiration; releases energy (ATP); many in sperm for swimming
ribosomes	make proteins
bacterial cell	cell wall, no nucleus (loop of DNA), plasmids, flagellum, slime capsule

MOVEMENT ACROSS MEMBRANES

Diffusion: net movement of molecules from high to low concentration (down a gradient); passive.

Active transport: against a concentration gradient (low to high), using energy from respiration (ATP), via carrier proteins.

Osmosis: movement of water from high to low water potential across a partially permeable membrane.

SA:V (Jun 24): a 3 cm cube has SA 54 cm^2, volume 27 cm^3, ratio 2 : 1. Smaller SA:V (emphysema, Jun 19) means less efficient gas exchange.

MODEL ANSWER · EFFECT OF TEMPERATURE ON ENZYME ACTIVITY (4–6)

Low temperature: less kinetic energy, fewer collisions between enzyme and substrate, fewer enzyme–substrate complexes, slower rate.

Above the optimum: the enzyme is denatured; the shape of the active site changes; the substrate no longer fits; fewer or no enzyme–substrate complexes; the rate falls.

pH: away from the optimum the active site changes shape in the same way (Nov 21: not at optimum pH).

MODEL METHOD · ENZYME AND TEMPERATURE (6) · JUN 24, JUN 26

Put a set volume and concentration of substrate (e.g. hydrogen peroxide) into a boiling tube and the enzyme (catalase) into another; place both in a water bath at the chosen temperature until they equilibrate; mix; measure the volume of oxygen with a gas syringe (or count bubbles) at set time intervals; repeat at other temperatures; keep volume and concentration the same; repeat each temperature and calculate a mean.

- Rate from a graph (Jun 26): change ÷ time, converted to the units asked: 0.5 AU ÷ 96 min = 0.0052 AU per minute.
- Why the rate slows over time (Jun 23): substrate is used up, so fewer enzyme–substrate complexes form.
- Immobilised enzymes (Nov 20, Jun 22, Jan 23): reusable, product not contaminated by enzyme, more stable to temperature/pH. Beads: sodium alginate + enzyme dropped into calcium chloride, then washed.

4 NUTRITION, DIGESTION AND FOOD TESTS

Test	Method	Positive result
reducing sugar	add Benedict's solution; heat in a water bath	blue → brick red (green/yellow/orange for less)
starch	add iodine solution	brown/orange → blue-black
protein	add biuret solution ($NaOH + CuSO_4$)	blue → lilac/purple
lipid	shake with ethanol, then add water	cloudy white emulsion

ENERGY IN FOOD · NOV 20, JUN 21, JUN 22, NOV 24

Method: weigh the food; measure a set volume of water into a boiling tube and take its starting temperature; ignite the food on a mounted needle and hold it under the tube until it is burnt; record the final temperature; repeat for each food and compare.

Energy (J) = mass of water (g) × 4.2 × temperature rise: 28 × 12 × 4.2 = 1411 J (Nov 20).

Errors: heat lost to the air, tube and needle; incomplete burning; uneven temperature (no stirring). Improvement: calorimeter with oxygen supply, lid, stirrer.

Energy from composition (Jan 22): 1.2 g protein × 17 kJ + 4.8 g lipid × 37 kJ = 198 kJ.

- Balanced diet: all nutrients in the correct amounts. Deficiencies: vitamin C → scurvy; iron → haemoglobin; vitamin D → calcium absorption; vitamin A → eye pigment (night vision); calcium → bones.
- Bile (Jan 23): made in the liver, stored in the gall bladder; emulsifies fat to increase the surface area for lipase; neutralises stomach acid.
- Peristalsis (5, Jan 23): circular muscles contract behind the bolus and relax in front, pushing food along; fibre adds bulk and stimulates the contractions.
- Villi: large surface area (villi and microvilli), thin wall, capillaries maintain the concentration gradient.
- BMI = mass (kg) ÷ height^2 (m^2): 67.0 ÷ 1.69^2 = 23.5, healthy (Nov 24).

5 BLOOD, CIRCULATION AND TISSUE FLUID

Component	Function
red blood cell	haemoglobin carries oxygen; no nucleus; biconcave
phagocyte	engulfs and digests pathogens (phagocytosis)
lymphocyte	produces antibodies (and antitoxins)
platelet	clotting
plasma	~90% water; transports dissolved nutrients, CO_2, urea, hormones, heat

MODEL ANSWER · BLOOD CLOTTING (5–6) · NOV 21, NOV 23

Damaged platelets release an enzyme (thromboplastin); prothrombin is converted to thrombin; thrombin converts soluble fibrinogen to insoluble fibrin; fibrin forms a mesh that traps red blood cells to form a clot; calcium ions are needed. The clot prevents blood loss and the entry of pathogens, preventing infection.

HEART AND VESSELS

Right side: deoxygenated blood to the lungs (pulmonary artery). Left side: oxygenated blood to the body (aorta). The left ventricle wall is thickest because it pumps blood at high pressure around the whole body.

Valves prevent backflow (e.g. from ventricle to atrium when the ventricle contracts). Arteries carry blood away from the heart, veins towards it; capillaries allow exchange by diffusion.

Blood pressure (Jun 26): highest in the aorta (contraction of the left ventricle); arteries recoil due to elastic tissue; pressure falls through capillaries (no elastic tissue, large total cross-section) giving a smooth, non-pulsatile flow suited to exchange.

TISSUE FLUID · NOV 24, JUN 25

High hydrostatic pressure at the arterial end forces water and small molecules out through gaps in the capillary wall; large plasma proteins stay in.

Jun 25: arterial end 4 − 3.3 = 0.7 kPa net outwards; venous end 1.6 − 3.3 = −1.7 kPa net inwards. Excess fluid drains into lymph vessels. Leaky capillaries: blood volume and pressure fall.

- Coronary heart disease (Nov 24): fatty deposits (atheroma) in coronary arteries reduce blood flow, so heart muscle lacks oxygen and glucose for respiration.
- Heart rate and exercise method (Jun 26): resting pulse at wrist/neck for 60 s; exercise for a set time; measure again; repeat and average.
- Speed calculation (Nov 25): 260 mm ÷ 0.25 s = 1040 mm/s = 1.04×10^3 mm/s.

6 BREATHING AND GAS EXCHANGE

MODEL ANSWER · INHALATION (4–6) · JUN 24, JUN 25, NOV 25

The external intercostal muscles contract, moving the ribs up and out; the diaphragm contracts and flattens; the volume of the thorax increases; the pressure inside the lungs decreases below atmospheric pressure; air is drawn in through the trachea.

- Alveoli adaptations: large surface area; thin walls (one cell thick); good blood supply (capillaries); moist surface. Emphysema: walls break down, surface area falls, less gas exchange.
- Lung volumes (Nov 21, Jun 24): tidal volume (normal breath), vital capacity (deepest breath in then out), residual volume (left after maximum exhalation).
- Control (Jun 23): more CO_2 in the blood is detected by chemoreceptors, so breathing rate and depth increase to remove it.
- Limewater method (Nov 21): breathe out through one tube and in through the other; exhaled air turns limewater cloudy faster because it contains more CO_2 from respiration.
- Smoking (Jun 19): tar paralyses cilia, mucus builds up, more infections and coughing.
- Blood flow in exercise (Nov 23): large increase to skeletal muscles and skin, decrease to the gut; more blood to the skin loses heat generated by respiration.

7 EXCRETION: LIVER, KIDNEY, DIALYSIS

THE NEPHRON

Ultrafiltration (glomerulus → Bowman's capsule): high blood pressure, because the afferent arteriole is wider than the efferent, forces water, glucose, urea and salts out; blood cells and proteins are too large.

Selective reabsorption (proximal tubule): all glucose returns to the blood by active transport.

Water reabsorption (collecting duct): by osmosis, controlled by ADH.

MODEL ANSWER · HOT DAY OR DEHYDRATION (5) · JUN 22, NOV 23, JUN 26

More sweating reduces the water potential of the blood; this is detected by the hypothalamus; the pituitary gland releases more ADH; ADH makes the collecting ducts more permeable; more water is reabsorbed into the blood; a small volume of concentrated urine is produced.

Alcohol reduces ADH (Jun 25): collecting ducts less permeable, less water reabsorbed, a large volume of dilute urine.

MODEL ANSWER · DIALYSIS (4) · JAN 20

Dialysis fluid contains no urea (and normal glucose and salt levels), so urea diffuses out of the blood across the partially permeable membrane down a concentration gradient. Heparin is added to prevent clotting in the machine. A transplant is permanent and allows a normal diet, but risks rejection.

- Liver: deamination of excess amino acids forms urea; detoxifies alcohol and hydrogen peroxide; makes bile; stores glycogen (Jun 26).
- Protein in urine (Jun 26) suggests kidney damage or high blood pressure; test with biuret.

- Renal artery vs renal vein (Jan 20): the artery has more oxygen, urea and salts.

8 NERVOUS SYSTEM AND SENSES

MODEL ANSWER · SYNAPSE (6) · JUN 21, NOV 25

The impulse arrives at the end of the axon (synaptic knob); vesicles release a neurotransmitter (e.g. acetylcholine) from the pre-synaptic membrane; it diffuses across the synaptic cleft; it binds to receptors on the post-synaptic membrane; this triggers a new impulse in the next neurone.

	Sensory neurone	**Motor neurone**
direction	from receptor to CNS	from CNS to effector (muscle or gland)
cell body	in the middle/side, outside the CNS	at one end, inside the CNS
axon	short	long

- Reflex action: rapid, automatic, protective, no conscious brain involvement. Arc: receptor → sensory → relay → motor → effector. Myelin sheath insulates and speeds the impulse.
- Nerve speed (Jun 26): 0.25 m ÷ 0.0045 s = 55.6 m/s. Hormones are slower because they travel in the blood.
- Accommodation for a near object (Jun 23): ciliary muscles contract, suspensory ligaments slacken, the lens becomes thicker (more convex) and refracts light more.
- Pupil in bright light: circular muscles of the iris contract, radial relax, pupil narrows. Method (Nov 24): measure pupil diameter in a dark room, shine a light, measure again.
- Defects: short sight (image in front of retina) – concave lens; long sight (behind retina) – convex lens; cataract – cloudy lens, replaced by surgery.
- Ear (Jan 23, Nov 25): ossicles transmit and amplify vibrations; cochlea hair cells generate impulses; Eustachian tube equalises pressure; semicircular canals detect head movement for balance.

9 HORMONES AND HOMEOSTASIS

MODEL ANSWER · HOMEOSTASIS DEFINITION (3) · JAN 22, JUN 25

The maintenance of a constant internal environment despite changes in the external environment, by negative feedback.

MODEL ANSWER · OVERHEATING (6) · JUN 24, JUN 25

The hypothalamus detects the rise in blood temperature; arterioles in the skin dilate (vasodilation), so more blood flows near the surface and heat is lost by radiation; sweat glands produce more sweat, which evaporates using heat energy from the skin.

Cold (Jan 19): shivering releases heat from respiration; vasoconstriction reduces blood flow to the skin; less sweating; hairs stand up to trap an insulating layer of air.

Why too high a temperature is dangerous (4): enzymes are denatured, the active site changes shape, the substrate cannot bind, so metabolic reactions slow or stop.

BLOOD GLUCOSE

High glucose: the pancreas releases insulin; the liver and muscles take up glucose and convert it to glycogen. Low glucose: glucagon converts glycogen back to glucose. Feedback stops secretion when the level is normal.

Diabetes (Nov 23): test urine with Benedict's or a glucose test stick; high blood glucose lowers water potential, water leaves cells by osmosis and is lost in urine (Jan 20).

- Adrenaline (Jan 20): increases heart rate and breathing, dilates airways, converts glycogen to glucose, redirects blood to muscles. Beta-blockers block its receptors, so the heart rate is lower.
- Matching table (Nov 24): ADH – water level of blood; thyroxine – metabolic rate; adrenaline – reduces blood flow to the gut; cerebrum – conscious thought; semicircular canals – body position; liver – urea formation.

10 SKELETON AND MUSCLES

- Synovial joint parts: cartilage (smooth, reduces friction, absorbs shock), synovial membrane (secretes fluid), synovial fluid (lubricates), ligaments (hold bones together), capsule. Elbow and knee: hinge; hip and shoulder: ball and socket.
- Raising the forearm (Jun 25, Nov 24): biceps contracts, triceps relaxes; they are an antagonistic pair; inelastic tendons transmit the pull to the radius.
- Muscle types (Nov 24): cardiac – striated, branched, single nucleus; skeletal – striated, unbranched, many nuclei; smooth – non-striated, spindle-shaped, single nucleus (gut, arteries, iris).
- Bone (Nov 25): compact bone, marrow, cartilage; needs calcium, phosphate, protein and vitamin D.

11 REPRODUCTION

MENSTRUAL CYCLE · JAN 20

FSH from the pituitary makes a follicle develop in the ovary; the ovary secretes **oestrogen**, which repairs the uterus **lining** after menstruation; once a month an egg is released (**ovulation**); progesterone maintains the lining; if there is no fertilisation, progesterone falls and menstruation follows.

Pregnancy needs extra calcium for the growth of the fetus's bones. Breastfeeding passes antibodies to the baby and gives the correct balance of nutrients.

SPERM AND MEIOSIS · JUN 21, JAN 23

Sperm: tail for swimming; many mitochondria for aerobic respiration, releasing energy for movement; acrosome contains enzymes that digest the egg membrane; haploid nucleus.

Importance of meiosis (3): produces haploid gametes (23 chromosomes), so the diploid number (46) is restored at fertilisation and does not double; it also introduces genetic variation.

Sex determination: XX × XY gives XX : XY = 1 : 1, so a 50% chance of each sex.

MODEL ANSWER · VASECTOMY (4 + 3) · JAN 22

The vas deferens is cut, so sperm cannot pass out of the body; they cannot reach the egg in the oviduct, so fertilisation cannot occur. It does not stop HIV, because fluid from the prostate gland is still released and can contain the virus.

- Chromosome aberrations (Jan 22): the probability rises sharply in mothers over 35; Down's syndrome has an extra chromosome 21 (47 in total).

12 DNA, CELL DIVISION AND INHERITANCE

MODEL ANSWER · DNA STRUCTURE (4) · JUN 24, JUN 25

Two strands twisted into a double helix; made of nucleotides (phosphate, deoxyribose sugar, base); four bases; complementary base pairing A–T and C–G; bases held together by hydrogen bonds.

DNA vs RNA: double vs single strand; deoxyribose vs ribose; thymine vs uracil. DNA is in the nucleus (chromosomes).

	Mitosis	Meiosis
cells produced	two, genetically identical	four, genetically different
chromosome number	diploid (46)	haploid (23)
use	growth, repair, asexual reproduction	gametes; diploid restored at fertilisation; variation

- Skin cancer (Nov 21): UV light causes mutations in the layer of actively dividing skin cells, which can then divide uncontrollably (melanoma).
- Mutation (Jan 22, Jan 23, Nov 23): a change in the base sequence changes the amino acid sequence, so a different protein with a different shape forms (e.g. an enzyme whose active site no longer fits). Causes: UV, X-rays.
- Blood groups: I^A and I^B are codominant, I^o recessive. Group O has no antigens. Mismatched blood causes agglutination, which blocks vessels.
- Sex-linked (Jun 19, Jun 24, Jun 26): carried on the X chromosome; males affected, females carriers. $X^DX^d \times X^DY$ gives X^DX^D, X^DX^d, X^DY, X^dY: probability of an affected child 1/4.
- Pedigrees: recessive if unaffected parents have an affected child (skips generations); sex-linked if only males are affected and females are carriers.
- Genetic engineering of insulin (6, Jun 22): remove the plasmid; cut out the human insulin gene and cut the plasmid with restriction enzymes; join with ligase to form recombinant DNA; return to the bacterium; grow in a fermenter.
- Stem cells (Jun 22): embryonic – can form most cell types but raise ethical issues and tumour risk; adult – less rejection but limited differentiation.

13 MICROORGANISMS, DISEASE AND IMMUNITY

MODEL ANSWER · VACCINATION (4–5) · JUN 21

A weakened, dead or attenuated pathogen (with its antigens) is injected; lymphocytes produce antibodies; memory cells are formed; on later infection the memory cells produce antibodies quickly, so the person does not become ill. Herd immunity reduces transmission.

MODEL ANSWER · ANTIBIOTIC RESISTANCE (5) · JAN 23

Bacteria mutate; a resistant bacterium survives the antibiotic while others die; it reproduces and passes on resistance; overuse or not finishing the course speeds this up; a new antibiotic is then needed. Antibiotics cannot treat viruses such as HIV because viruses have no cell wall or metabolism to target.

- Phagocytosis: engulfs the pathogen, lysosome enzymes digest it.
- Malaria (Jun 19): the mosquito is the vector; it sucks infected blood and transfers the parasite on the next bite.
- Bacterial growth curve (Jun 24): lag, exponential (log), stationary (births equal deaths), death.
- Aseptic technique (6, Jun 19): wash hands and benches; flame the inoculating loop; lift the lid only slightly; tape the lid, not fully sealed; incubate at 25 °C or below to avoid growing pathogens; sterilise equipment.
- Antibiotic effectiveness (Jun 19): measure the diameter of the clear zone; larger zone, more effective.
- Sewage and sanitation (Jun 23, Jun 24): screens remove large waste; aerobic bacteria need oxygen; methane from digestion; pit latrines away from wells, covered to keep flies out.
- HIV and AIDS (Jun 26): HIV is the virus; AIDS is the disease it causes.

14 PRACTICAL SKILLS AND CALCULATIONS

Calculation	Method	Past-paper example
magnification	image size ÷ actual size, same units	64 mm = 64 000 µm; 64 000 ÷ 2.0 = ×32 000 (Jun 25)
actual size	image size ÷ magnification	20 mm ÷ 750 = 0.0267 mm = 26.7 µm (Nov 21)
percentage change	(new − old) ÷ old × 100	(2629 − 2103) ÷ 2103 × 100 = 25% (Jun 22)
percentage increase	same, then round as asked	(17 500 − 5600) ÷ 5600 × 100 = 212.5 = 210% to 2 s.f. (Nov 23)
ratio 1 : n	divide both by the first value	7500 : 13 750 = 1 : 1.83 (Nov 24)
mean	sum ÷ number	(175.9 + 178.5 + 166.3 + 167.6) ÷ 4 = 172.1 (Jun 19)
rate	change ÷ time	0.5 ÷ 96 = 0.0052 AU/min (Jun 26)
speed	distance ÷ time	0.25 ÷ 0.0045 = 55.6 m/s (Jun 26)
standard form	$a \times 10^n$, $1 \le a < 10$	70 000 000 ÷ 5000 = 1.4×10^4 (Jun 26)
scaling a rate	rate × population ÷ base	2.75 per 100 000 × 3 450 000 = 95 (Jun 21)

- Planning: independent variable (with a range), dependent variable (how measured), control variables (named), repeats and mean, safety with a reason. Improve reliability by repeating; improve validity by controlling variables.
- Results tables: headings with units, independent variable in the first column.
- Evaluate data: comment on sample size, representativeness and other variables (diet, exercise, age, sex).

15 FINAL A* CHECKLIST

1. Count the marks; make one creditworthy point per mark and add one spare.
2. Use the key terms: active site, enzyme–substrate complex, water potential, partially permeable, negative feedback.
3. Methods: measure, control (named), repeat, mean, safety with reason. Never “amount”.
4. Graphs: units, scale over half the grid, accurate plots, the line type asked, a key.
5. Data: describe with figures and units; then explain with biology.
6. Calculations: convert units first; show working; correct decimal places and units.
7. Genetics: full diagram with symbols, gametes, offspring, phenotypes, probability.
8. Name structures precisely: left/right, tendon/ligament, artery/vein, sensory/motor.
9. For “explain”: link every point with “so” or “because”.
10. Leave 10 minutes to check calculations and long answers.